Protein domains of human PMM2 (InterPro analysis)

What is a protein domain?

Proteins are rarely a single block. Most are built from regions called domains, each of which folds on its own and usually has a function to do, such as holding a substrate or carrying out a chemical reaction. Knowing where a variant sits relative to these regions is a first step towards understanding which function it may disturb. Related proteins are also grouped into families, which share ancestry and function, and homologous superfamilies, which share the same overall fold. InterPro is a database that combines several resources to recognize these families and domains in a protein sequence and to report which residues they cover.

What we did:

We entered the sequence of human PMM2 (UniProt O15305, 246 amino acids) into InterPro. For each of the five entries returned, we noted the accession, the type (family, F, or homologous superfamily, H) and the residues covered. We then checked whether position 237, the residue changed in the T237M variant, falls inside each one.

What it means:

Three of the entries (the phosphomannomutase family and the two HAD superfamilies) cover almost the whole protein, so it is no surprise that position 237 falls inside them. They simply confirm that PMM2 is a phosphomannomutase belonging to the haloacid dehalogenase (HAD) superfamily. The more informative entries are the specific ones: position 237 lies outside both the HAD subfamily IIB region (residues 9–225) and the cap domain (residues 86–187), near the C-terminal end of the protein.

Why this matters for the project:

Our alignments identified the variant at the sequence level (c.710C>T, p.Thr237Met), and the structural comparison asks whether it changes the shape of the protein. The domain analysis connects the two by showing where in the protein the change occurs.

It also helps us narrow down how the variant might act. The core and cap domains move relative to each other so that the enzyme can close around its substrate. Since position 237 is outside the cap domain, the effect of T237M is unlikely to come from a defect of this domain. This fits with the location of Thr237 in the α9 helix of the core domain (Briso-Montiano 2022).

Finally, it gives context to the conservation results. Thr237 is conserved in *Leishmania mexicana*, and its hydrogen-bond partner, Thr16, sits within the FDVDGTL stretch (residues 11–17) that is identical in all three species. So two positions that are far apart in the sequence come together in the folded protein, which supports the idea that position 237 matters.